LECTURE 1 Supplement

Topics

Stock market indexes and applications

- Dow Jones Industrial Average (DJIA) is an index that tracks 30 large well-established U.S. corporations
- The composition of the index changes on occasion to reflect emerging sectors in the economy
- The index is calculated using a *price-weighted average* methodology
- Suppose the stock prices of the 30 companies in the index today are: $P_1, P_2, ..., P_{30}$
- Then the value of the price-weighted average index today is given by:

$$I^{DJIA} = \frac{P_1 + P_2 + ... + P_{30}}{d}$$

- The *divisor d* is a known parameter that can change over time
- The divisor was 30 in 1928, but is around 0.152 today

- The DJIA divisor, *d*, changes in one of the following events:
  - A company in the index experiences a *stock split*:

    1 old share becomes N new shares

  - A company in the index experiences a *reverse stock split*:

    M old share becomes 1 new share

  - An old company is replaced with a new company

- In each of these events, the divisor is recalculated such that the value of the index *remains unchanged* on the date of the event

- Standard and Poor's Composite 500 (S&P500) is an index that tracks 500 of the largest U.S. corporations, by market capitalization (market value)
- The composition of the index changes when new firms that achieved high market capitalizations are adopted in the index
- The index is calculated using a *value-weighted average*
- Suppose the market value of the 500 companies in the index today are: $MV_1, MV_2, ..., MV_{500}$
- Then the value of the value-weighted average index today is:

$$I^{SP500} = \frac{MV_1 + MV_2 + ... + MV_{500}}{d}$$

- The *divisor* d is a known parameter that can change over time

- The market value, MV, is calculated as the product of stock price, P, and shares outstanding, N:

$$\mathrm{MV} = P \times N$$

- Just like in the case of DJIA, the divisor for S&P500, d, changes in one of the following events:
  - A company in the index experiences a stock split
  - A company in the index experiences a reverse stock split
  - An old company is replaced with a new company

- In each of these events, the divisor is recalculated such that the value of the index *remains unchanged* on the date of the event

Application 1

- A price-weighted index consists of 5 stocks and their prices and shares outstanding are in the table below:

Stocks	N (mil)	P 2023	P 2024
Stock 1	20	50	45
Stock 2	30	120	100
Stock 3	40	250	280
Stock 4	10	310	350
Stock 5	80	180	220

- If the index value was 1300 in 2023, what should be the value of the index in 2024?
- Calculate the relative change in the index from 2023 to 2024
- In 2024 the index dropped Stock 5 and added Stock 6, that was trading at $75 a share. What should be the new divisor in 2024?

Application 2

- A price-weighted index consists of 5 stocks and their prices and shares outstanding are in the table below:

Stocks	N 2024	N 2025	P 2024	P 2025
Stock 1	20	20	60	70
Stock 2	30	30	120	100
Stock 3	40	200	250	46
Stock 4	10	10	310	330
Stock 5	80	80	160	150

- Suppose the index had a value of 1450 in 2024
- Suppose further that Stock 3 had a 5-for-1 split at the end of 2024, right after the index was calculated
- What should be the value of the index in 2025?

Application 3

- A value-weighted index consists of 4 stocks and their prices and shares outstanding are in the table below:

Stocks	N 2024	N 2025	P 2024	P 2025
Stock A	25	25	60	58
Stock B	35	35	130	95
Stock C	45	45	80	105
Stock D	15	15	37	48

- If the index value was 1000 in 2024, what should be the value of the index in 2025?